Storage capacity automation for EVS and OBS in T Public Cloud

As our customer base scales across four distinct **T Public Cloud** regions and nine Availability Zones, our current reliance on manual storage management has transformed from a routine task into a critical operational bottleneck. The complex nature of manual **EVS** and **OBS** pool expansion leaves us dangerously exposed to sudden data surges and potential downtime, while a lack of intelligent orchestration has created a costly imbalance—wasting OpEx through over-provisioning in our smaller **Preprod** and **Amsterdam** environments while Production faces the constant stress of "near-capacity" alerts. Furthermore, the specialized needs of our "Dedicated" customers in the Hybrid region introduce a layer of regulatory complexity that traditional manual scripts simply cannot navigate, making a transition to automated, compliant governance no longer optional, but essential for our continued growth. **IBM** solutions could help us to govern these complex systems in one automated dashboard.

Region	Mission	Automation Behavior
Production (3 AZs)	**Performance**	**Aggressive Scaling:** Prioritize uptime. Auto-expand EVS pools at 75% threshold.
Amsterdam (3 AZs)	**Efficiency**	**Load Balancing:** Shift workloads between AZs to avoid purchasing new hardware.
Preprod (2 AZs)	**Cost Savings**	**Right-Sizing:** Automatically delete unattached snapshots and shrink underutilized volumes.
Hybrid (2 AZs)	**Compliance**	**Governed Growth:** Automation is "Recommendation-only" until a human clicks 'Approve' in Concert.

Fort he solution i will use all four IBM service:

Instana

Real-Time Precision Monitoring IBM Instana acts as the **"Sensory Layer"** for the entire storage fabric. While standard monitoring tools provide snapshots of data every few minutes, Instana captures **EVS (Block)** and **OBS (Object)** telemetry with **one-second granularity** across all 9 Availability Zones. By integrating directly with **Huawei Cloud Eye APIs.**

Metrics for EVS:

Metric ID	Metric Name	Purpose for Automation
disk_util_inband	**Disk Usage (%)**	**The Primary Trigger:** When this exceeds a threshold (e.g., 80%), it signals Turbonomic to initiate an EVS expansion.
disk_read_bytes_rate	**Read Bandwidth**	**Performance Check:** Used to determine if a volume should be moved to a high-performance SSD pool.
disk_write_bytes_rate	**Write Bandwidth**	**Growth Vector:** High write rates suggest a data surge that requires immediate capacity buffer.
disk_read_requests_rate	**Read IOPS**	**Saturation Level:** Detects if a pool is congested, triggering a "Move" action to a different AZ.
disk_device_latency	**IO Latency**	**Health Metric:** High latency despite low usage suggests underlying pool issues in that specific zone.

Metrics for OBS:

Metric ID	Metric Name	Purpose for Automation
stored_size	**Storage Space**	**Capacity Tracking:** Measures total bytes stored in a bucket; used for budget-based scaling alerts.
object_count	**Total Objects**	**Metadata Pressure:** Identifies buckets with millions of small objects that may need special indexing or lifecycle policies.
put_request_count	**Write Requests**	**Ingestion Load:** Tracks how fast data is arriving. A spike here predicts a capacity crunch in the next hour.
upstream_bandwidth	**Outbound Traffic**	**Cost Control:** Monitors egress costs, vital for **Preprod** to ensure testing doesn't exceed budget.

Turbonomic:

1. Scaling Actions (Dynamic Capacity)

- **Resize Volume (Scale Up) tcloud_scale_up_evs_capacity:** Automatically increases the capacity of a Huawei EVS disk (e.g., from 500GB to 1TB) when utilization exceeds a specific percentile (e.g., 90th percentile of usage).
- **Resize Volume (Scale Down) tcloud_scale_down_evs_capacity:** Identifies over-provisioned disks—common in **Preprod**—and recommends shrinking them to save costs without impacting the application.

- **IOPS/Throughput Adjustment tcloud_adjust_evs_performance_tier:** For EVS types that support it (like Ultra-high I/O), Turbonomic can adjust the provisioned performance limits independently of capacity to handle temporary traffic spikes.

2. Placement & Migration Actions (Load Balancing)

- **Storage Move (Intra-pool) tcloud_rebalance_evs_pool_placement:** Moves an EVS volume from a heavily congested storage pool in **pool 1** to a less-utilized pool in **pool 2** This balances the I/O load across the physical hardware.
- **Cross-Tier Migration tcloud_migrate_evs_storage_tier:** Automatically moves a workload from a high-cost storage tier (SSD/Ultra-high I/O) to a lower-cost tier (SATA/Common I/O) if the performance data shows the app doesn't require high speed and customer approved it.
- **Affinity/Anti-Affinity Enforcement tcloud_enforce_storage_anti_affinity:** Ensures that critical redundant volumes (like a primary and standby database) are never placed on the same physical storage hardware, even as the environment scales.

3. Savings & Hygiene Actions (Waste Reclamation)

- **Delete Unattached Volumes tcloud_purge_unattached_evs_volumes:** Identifies EVS disks that are no longer attached to any ECS instance (often left behind after a test in **Preprod**) and executes a "Safe Delete" (snapshot + delete).
- **Delete Old Snapshots tcloud_cleanup_expired_evs_snapshots:** Identifies snapshots that are older than the company's retention policy and deletes them.
- **OBS Lifecycle Management tcloud_execute_obs_lifecycle_transition:** Automatically moves objects from **Standard OBS** to **Infrequent Access** or **Cold Archive** based on the frequency of access detected by Instana.

4. Provisioning & Prevention Actions (Growth Readiness)

- **Provision New Storage Pool tcloud_provision_storage_pool_capacity:** When total regional capacity reaches a dangerous threshold (e.g., 85%), Turbonomic generates a "Provision" action to trigger the deployment of new storage backend resources.
- **Pre-emptive Move tcloud_evacuate_at_risk_storage:** Moves a workload *before* a failure occurs. If a storage controller shows signs of degradation (high latency), Turbonomic evacuates the volumes to a healthy zone.

Concert

This library of IBM Concert functions establishes the **Governance Circuit Breaker** necessary for truly autonomous storage operations across our global T Public Cloud footprint. By mapping raw infrastructure tasks to Concert's **six key dimensions**—specifically Resilience, Compliance, and Vulnerability—these functions transform standard API calls into useful transactions. Whether it is enforcing **physical geofencing** to maintain data sovereignty in the Hybrid region or conducting a **pre-expansion vulnerability scan** via SBOM analysis, this logic layer ensures that the data shifted across our nine Availability Zones is audited, secure, and legally compliant. Ultimately, these functions provide the **Explainable AI (XAI)** reasoning trace required for senior oversight, proving that our automation doesn't just react to capacity issues, but actively hardens our resilience posture with actions it takes.

Concert Function Name	Dimension	Logical Operation	Required tcloud Input
concert_validate_resilience_posture	**Resilience**	Checks the application's current **Resilience Score** and **NFR targets** before allowing a resize.	region_id, volume_id
concert_check_compliance_control	**Compliance**	Cross-references the action against your **Compliance Profile** (e.g., SOC2, NIST) to ensure security controls remain intact.	compliance_profile_id
concert_verify_geofence_sovereignty	**Compliance**	**The Hard Stop:** Validates that data-movements stay within the specific	az_id, physical_site_id

Concert Function Name	Dimension	Logical Operation	Required tcloud Input
		Physical Site ID mapped in the inventory.	
concert_assess_vulnerability_risk	**Vulnerability**	Scans the "Software Bill of Materials" (**SBOM**) of the app using the storage to ensure no high-risk CVEs are present during expansion.	app_id, risk_score_threshold
concert_calculate_disruption_impact	**Resilience**	Estimates the **Disruption Cost** of a storage move (e.g., IOPS latency during migration) and decides if it exceeds the business limit.	estimated_downtime_mins
concert_audit_retention_policy	**Compliance**	Validates if a Delete or Archive action matches the legal **Data Lifecycle** requirements for that customer.	snapshot_age, policy_id
concert_trigger_approval_workflow	**Operations**	Routes high-impact actions (e.g., provisioning a new pool) to **ServiceNow** or **Slack** for human/AI-Agent sign-off.	ticket_system_id, approver_role

Concert Function Name	Dimension	Logical Operation	Required tcloud Input
concert_log_resilience_metric	**Resilience**	Updates the **Post-Action Posture** (e.g., new MTTR or Availability % after the storage was fixed).	mins_mttr, pct_availability
concert_ai_reasoning_trace	**Agentic AI**	(New 2026) Captures the **"Reasoning Trace"** of why the AI chose a specific storage fix for audit and "Explainable AI" logs.	execution_token_id

Watson

In this architecture, watsonx acts as the **Agentic Orchestrator**, bridging the gap between high-level business intent and the low-level execution of T Public Cloud storage APIs.

Skills library:

Watson Skill Name	Underlying Logic	Senior Outcome
skill_autonomous_expansion	Triggers: hcloud_scale_up_evs + concert_validate_resilience	**Hands-Free Growth:** The system grows itself without a ticket.
skill_automated_waste_reclamation	Triggers: hcloud_purge_unattache	**Instant Savings:** AI identifies and kills "Digital Waste" daily.

Watson Skill Name	Underlying Logic	Senior Outcome
	d + concert_audit_retention	
skill_sovereign_data_migration	Triggers: hcloud_rebalance_evs + concert_verify_geofence	**Risk-Free Mobility:** Data moves within zones but never crosses borders.
skill_preemptive_resilience_fix	Triggers: hcloud_evacuate_at_risk_storage + instana_health_check	**Ghost-in-the-Machine:** Fixes hardware issues before they become outages.
skill_natural_language_provisioning	Triggers: watsonx_code_gen + hcloud_provision_pool	**On-Demand Capacity:** Provision new pools via a simple chat command.

Standard Use-Case Workflows

Use-Case A: Production Capacity Surge (Performance Focus)

- **Trigger:** Instana detects a write-speed spike in **Production**.
- **Decision:** Turbonomic identifies an IOPS bottleneck.
- **Governance:** Concert confirms the customer has a "Premium" SLA and approves the change.
- **Action:** watsonx executes **tcloud_adjust_evs_performance_tier** to move the volume to Ultra-High I/O.

- **Outcome:** Zero latency for the customer; no manual intervention required.

Use-Case B: Preprod Waste Reclamation (Cost Focus)

- **Trigger:** Instana identifies 500GB of "Unattached" EVS volumes in **Preprod**.
- **Decision:** Turbonomic marks these as "Zombie" resources for deletion.
- **Governance:** Concert checks the **tcloud_cleanup_expired_snapshots** policy to ensure data retention laws (7 days) are met.
- **Action:** watsonx executes tcloud_purge_unattached_evs_volumes.
- **Outcome:** Immediate reduction in Huawei Cloud OpEx; cleaner testing environment.

Use-Case C: Hybrid Sovereignty Protection (Compliance Focus)

- **Trigger:** A storage pool in the **Hybrid Region** reaches 85% capacity.
- **Decision:** Turbonomic suggests moving data to a cheaper pool in the **Public Region**.
- **Governance: Concert blocks the move.** It detects a "Dedicated Hardware" constraint and triggers concert_trigger_approval_workflow instead.
- **Action:** watsonx waits for a human signature, then executes a local expansion (**tcloud_scale_up_evs**) within the same physical site.
- **Outcome:** 100% compliance with data residency laws; legal risk averted.